



GATE CS / IT 2027

Digital Logic

Quick Revision Sheet

Complete GATE syllabus coverage — All topics and subtopics

Number Systems

Conversions & Codes

- Binary to Decimal: use positional weights
- Hex to Binary: expand each hex digit to 4 bits
- BCD: 4 bits per decimal digit (0-9 only valid)
- Gray Code: $G[n] = B[n] \oplus B[n-1]$
- 1s Complement: invert all bits
- 2s Complement: invert all bits then add 1
- Range of n-bit 2s complement: $-2^{(n-1)}$ to $2^{(n-1)}-1$

Boolean Algebra

Laws

- Identity: $A+0=A$, $A.1=A$
- Null: $A+1=1$, $A.0=0$
- Idempotent: $A+A=A$, $A.A=A$
- Complement: $A+A=1$, $A.A=0$
- Absorption: $A+AB=A$, $A(A+B)=A$
- De Morgan: $(A+B)=AB$, $(AB)=A+B$

K-Map

- Groups must be powers of 2 (1,2,4,8...)
- Wrap-around allowed
- Largest groups give minimal expression
- SOP: group 1s | POS: group 0s
- Dont-care cells can be used in grouping

Combinational Circuits

Adders & Multiplexers

- Half Adder: $\text{Sum} = A \oplus B$, $\text{Carry} = A.B$
- Full Adder: $\text{Sum} = A \oplus B \oplus C_{in}$, $\text{Cout} = \text{majority}(A,B,C_{in})$
- 2-to-1 MUX: $Y = S.B + S.A$
- MUX can implement any boolean function
- $n:2^n$ Decoder, $2^n:n$ Encoder

Sequential Circuits

Flip-Flops

- SR: forbidden state $R=S=1$, $Q+ = S + R.Q$
- D: $Q+ = D$ (most common, no race condition)
- JK: $Q+ = J.Q + K.Q$ ($J=K=1$ toggles)
- T: $Q+ = T \oplus Q$ ($T=1$ toggles, $T=0$ holds)

FSM

- Mealy: output depends on state AND input
- Moore: output depends only on current state
- State minimization merges equivalent states
- Steps: state table $\rightarrow$ K-map $\rightarrow$ circuit

